Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Project Overview

Dataset

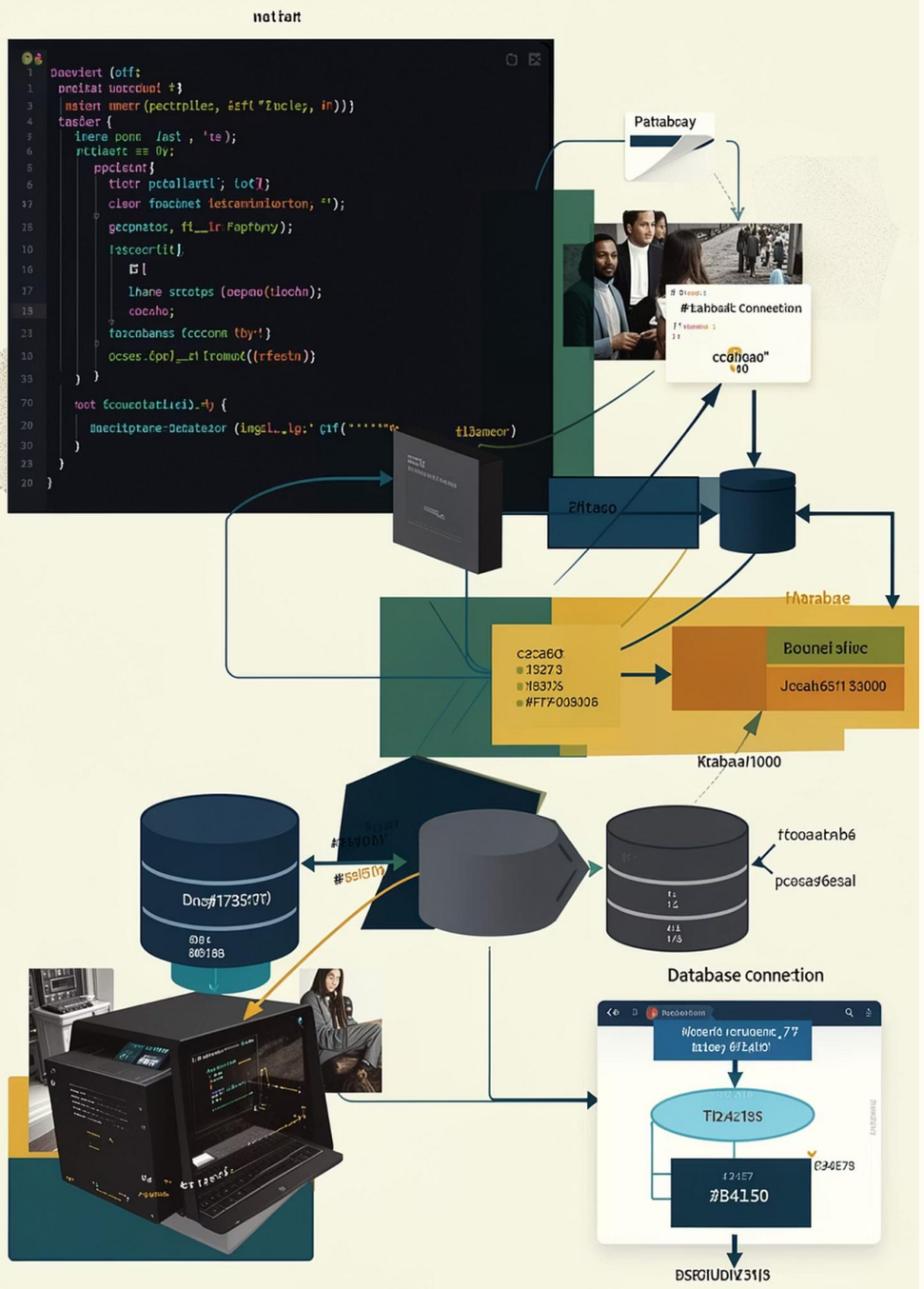
3,900 purchases across 18 columns with customer demographics, purchase details, and shopping behavior

Goal

Uncover spending patterns, customer segments, product preferences, and subscription behavior

Outcome

Strategic recommendations for boosting subscriptions, loyalty programs, and targeted marketing



Data Preparation Process

Python Analysis

- *Data loading with pandas*
- *Missing data imputation using median ratings*
- *Feature engineering: age groups and purchase frequency*
- *Column standardization to snake case*

SQL Integration

- *Connected to PostgreSQL database*
- *Loaded cleaned DataFrame for structured queries*
- *Performed 10 key business analyses*

Revenue by Gender

$157,890

Male Customers

Generated 68% of total revenue

$75,191

Female Customers

Generated 32% of total revenue

Male customers spent significantly more than female customers across all categories analyzed.

Top Products by Rating

1

Gloves

Average rating: 3.86

2

Sandals

Average rating: 3.84

3

Boots

Average rating: 3.82

4

Hat

Average rating: 3.80

5

Skirt

Average rating: 3.78

Shipping & Subscription Analysis

Shipping Comparison

Standard: *$58.46 average*

Express: *$60.48 average*

Express shipping customers spent 3.5% more per purchase.

Subscription Status

Subscribers (1,053): *$59.49 avg spend, $62,645 total revenue*

Non-subscribers (2,847): *$59.87 avg spend, $170,436 total revenue*

Discount-Dependent Products

50%

Hat

Half of all hat purchases used discounts

49.66%

Sneakers

Nearly 50% discount rate

49.07%

Coat

High discount dependency

48.17%

Sweater

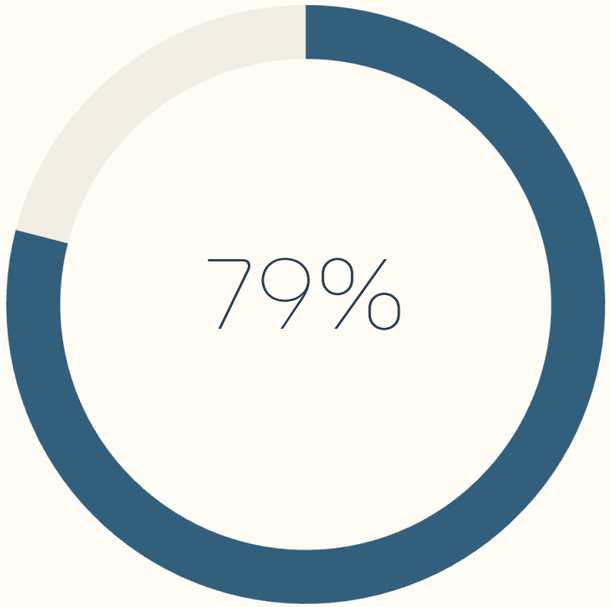
Strong discount influence

47.37%

Pants

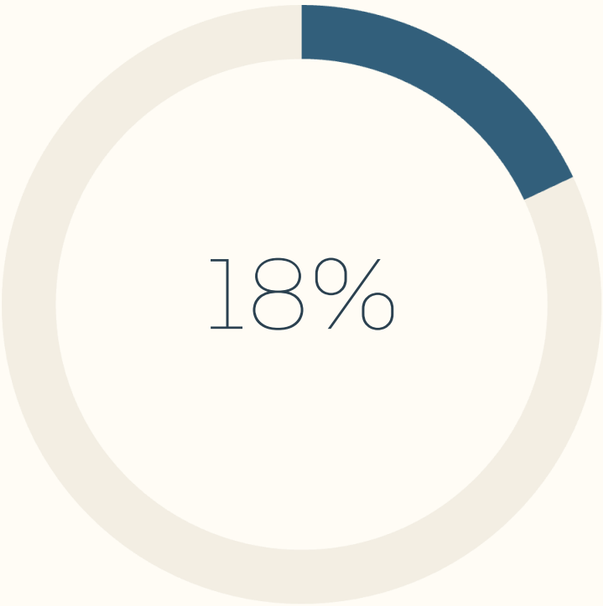
47% discount usage

Customer Segmentation



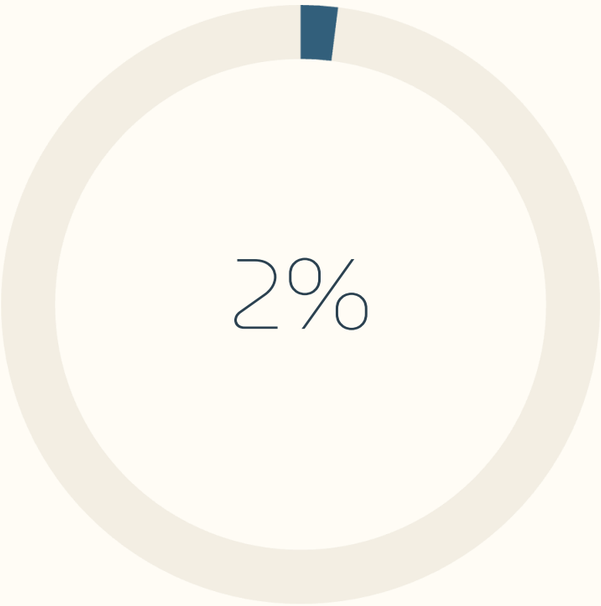
Loyal Customers

3,116 customers with consistent purchase history



Returning Customers

701 customers making occasional repeat purchases



New Customers

83 first-time buyers in the dataset

Revenue by Age Group

Young Adult

$62,143 total revenue

Middle-aged

$59,197 total revenue

Adult

$55,978 total revenue

Senior

$55,763 total revenue

Young adults generated the highest revenue, followed by middle-aged customers. Revenue distribution across age groups was relatively balanced.

Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits and incentives for subscribers to increase conversion rates

02

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment and increase retention

03

Review Discount Policy

Balance sales boosts with margin control, especially for high-discount products like hats and sneakers

04

Product Positioning

Highlight top-rated products (gloves, sandals, boots) in marketing campaigns

05

Targeted Marketing

Focus efforts on high-revenue age groups (young adults, middle-aged) and express-shipping users